

R525 Single-stage Reverse Osmosis

Control System Operation Manual



R525 Single-stage reverse osmosis controller, equipped with a 7-inch color high-definition LCD touch screen, the interface dynamically the system and the physical machine combined, fault issues and operation steps are clearly visible. The wiring is equipped with pluggable terminals, which are anti-reversal and anti-ose connection, and the maintenance is convenient. The card slot fixing machine is easy to install, and the wiring position of the accessories is clearly marked and easy to understand. It can extended to network and remote control functions, and the manufacturer information can be customized.

Functional introduction

1、Dry protection

When the raw water is interrupted and the water level of the raw water tank is low, the controller automatically stops the operation of the entire system to prevent the system idling, and the LCD display status prompt **Raw water not full**. The entire control system stops running. The controller constantly monitors the low-water switch, and when the water level returns to normal, the controller automatically restores system. (Figure 1)

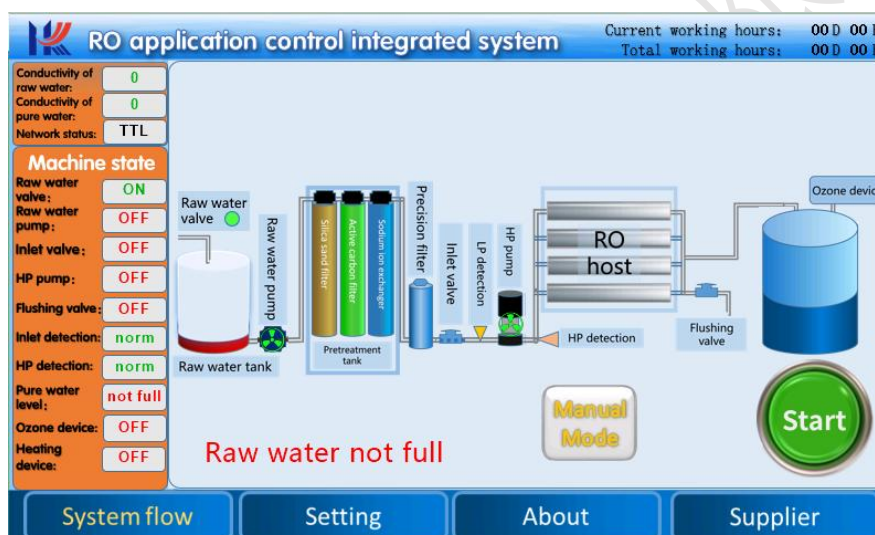


Figure 1

2、Low Voltage Protection

When the pretreatment system is in a state of water shortage or the low pressure switch is damaged, and cannot provide the RO system with normal water supply pressure (ufficient low pressure), the controller temporarily shuts down the water production system, LCD display status prompt **Low pressure fault delaye**. Display interface stops dynamic operation, fault is displayed at the low-voltage switch **red**. (Figure 2)

Thereafter, the controller will continuously detect the low pressure switch, and when the pressure returns to normal, the water control system will operate

normally after reaching the set.

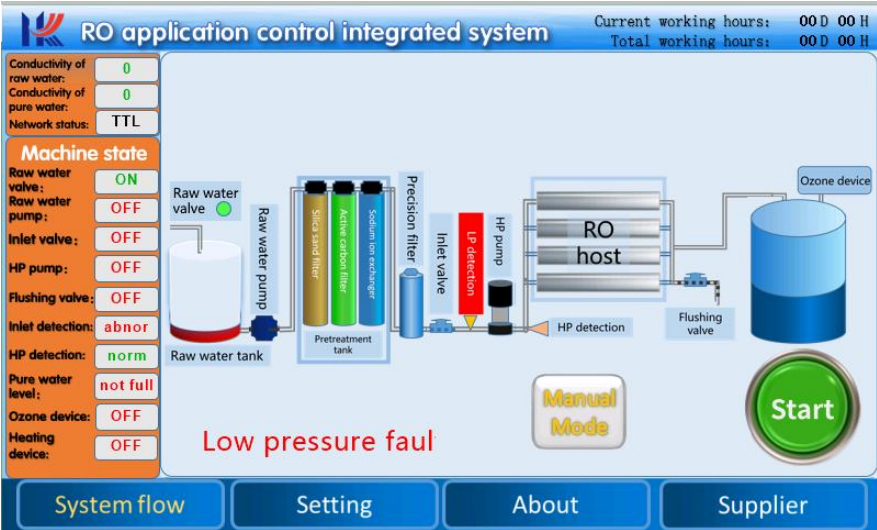


Figure 2

3、High Voltage Protection

When the pressure in front of the RO membrane is too high or the high-pressure switch is damaged, the controller shuts down the entire RO system, and LCD display shows the status prompt **High pressure fault delay**, High pressure switch shows fault **red**. (Figure 3)

Thereafter, the controller will continuously detect the high-pressure switch, and when the pressure returns to normal, the water production control system will operate normally after reaching set time.

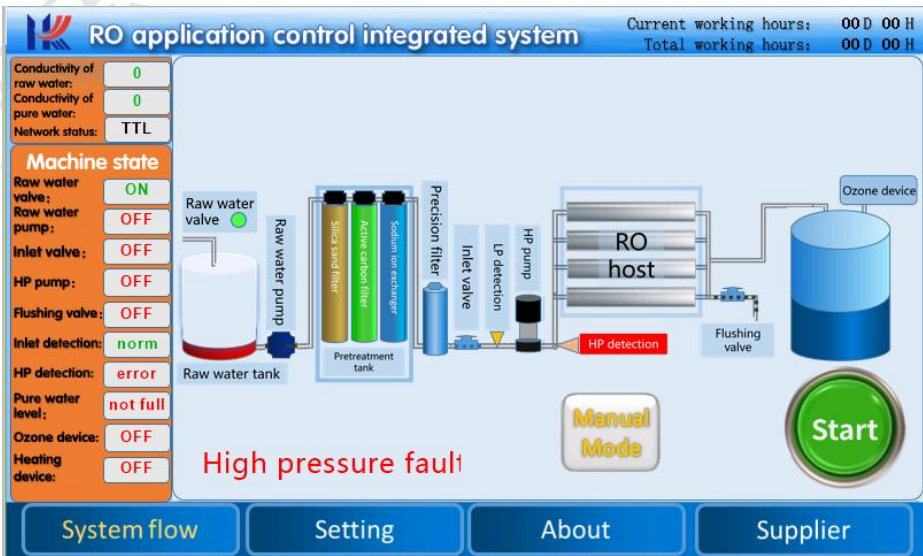
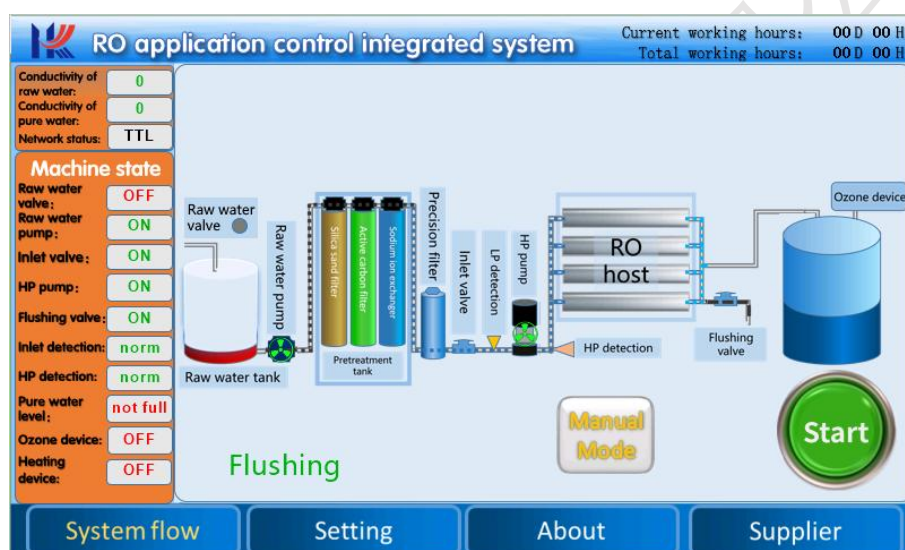


Figure 3

4、 Membrane flushing function

When the equipment is running normally, the system performs a first membrane flush each time it is turned on for operation, and during the water production process, the defaults to a flush of 30 seconds every 60 minutes (this parameter can be manually adjusted). After the pure water tank is full, a membrane flush is performed, and then every 2 hours in standby mode, a flush of 30 seconds is performed. (This parameter can be manually adjusted). (Figure 4)

Figure 4



5、 Feed water tank level control function

When the water level of the raw water tank is lower than the high-level float ball, the control system opens the raw water valve to fill water, and the water level of the raw water tank reaches the high water level, the control system will close the raw water valve. (Figure 5)

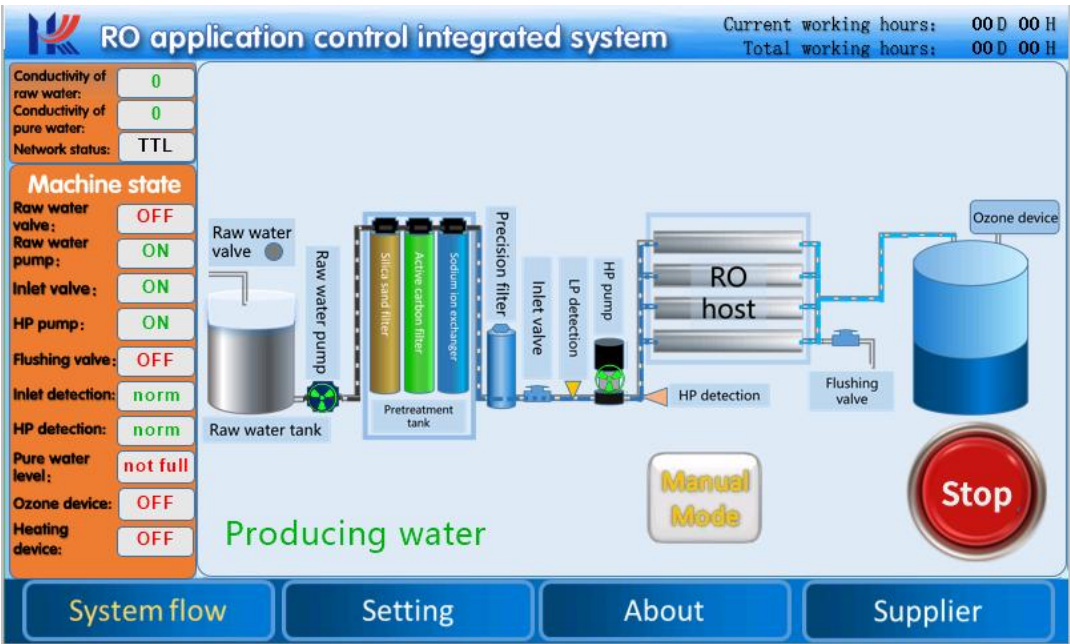


Figure 5

6、Pure water tank level control function

When the level of the pure water tank is lower than the high-level float ball, the control system will start the water production system to produce water, until level of the pure water tank reaches the preset high level, the system completes the membrane flush and automatically enters the standby state. (Figure 6)

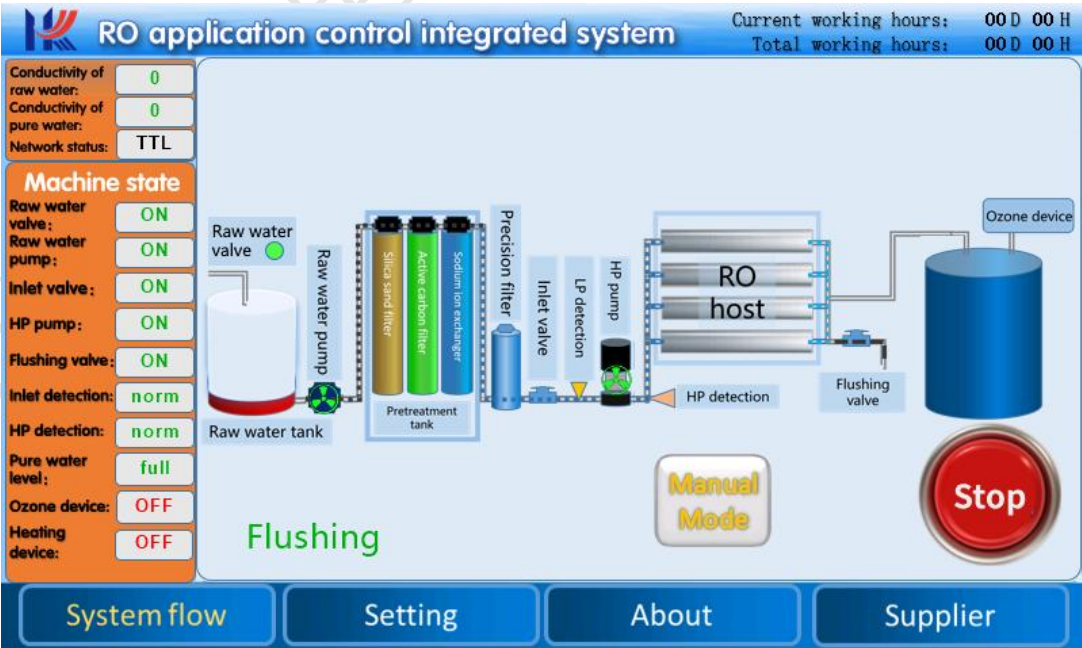


Figure 6

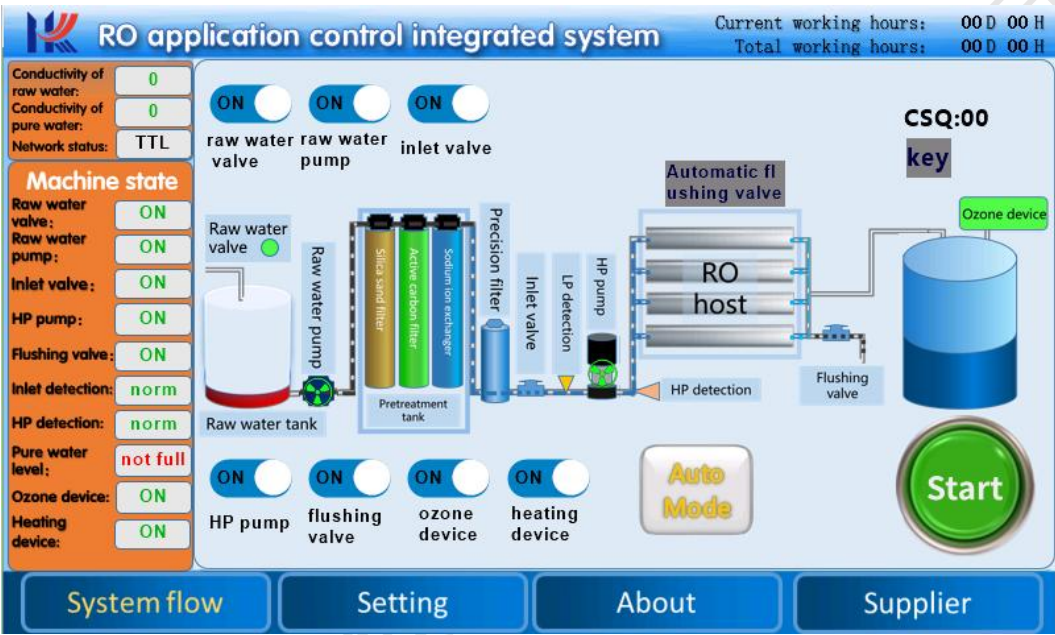
7. Auto Mode

Test if each output is normal, the button turns on manually  Status。

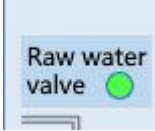
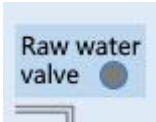
(Figure 7)

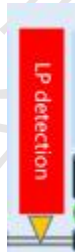
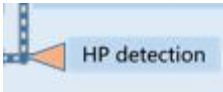
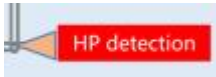
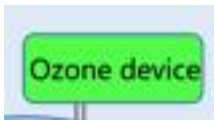
Note: When manual tuning is enabled, please press the buttons in order.

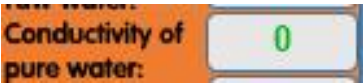
Figure 7



8、Screen on indicator and off state

Raw water valve operation indication	Raw water valve open 	Raw water valve closed 
Raw water pump operation indication	Raw water pump startup 	Raw water pump shutdown 

Inlet valve operation indication	Inlet valve open 	Inlet valve closed 
Low-voltage switch indicator	Low pressure normal 	Low pressure anomaly 
High pressure pump operation indicator	High pressure pump start 	High pressure pump off 
High pressure switch indicator	High pressure normal 	High voltage anomaly 
Flush valve operation indicator	Flush valve open 	Flush valve closed 
Ozone sterilization operation indicator	Ozone sterilization start 	Ozone sterilization off 

Source water quality testing		Detect the concentration of heavy metal ions in the raw water tank
Pure water quality testing		Detect the concentration of heavy metal ions in the pure water tank

Product parameters

1. Power supply voltage: DC24V;
2. Display: 7-inch LCD screen;
3. Touch type: Resist touch screen;
4. Backlight life: >20000 hours (Continuous operation at maximum brightness, half brightness time);
5. Consumption: $\geq 5W$;
6. Working environment: Temperature $-20-70^{\circ}C$ Humidity $\leq 85\% RH$;
7. Water quality detection: High precision dual-path conductivity;
8. Case size: 203mm*149mm*45mm;

Run parameters setting

1. Click the parameter setting button in the main interface to enter the setting interface;
2. The range and description of the running parameter setting are in the following figure;

RO application control integrated system Current working hours: 00 D 01 H
Total working hours: 00 D 02 H

flushing time in water making process	30 S	Flushing interval in water making process	60 M
Ozone sterilization time	30 S	Ozone sterilization interval	120 M
Flushing time after full water	30 S	Flushing interval after full water	2 H
Timeout time for a single continuous water production	4 H	Delay time for starting water production	0 M
Flush mode	HP flush	Delay time for low pressure detection	10 S
Alarm method when the temperature is too low	alarm	Delay time for restarting after device failure	5 M
Heater switch	ON	Starting temperature of temperature control heating	5 °C
Temperature control heating stop temperature	10 °C	Operating mode when power on	memory
Delayed opening time of HP pump after low pressure	5 S	Number of retries after low pressure abnormality	3

Click the Save button to save the parameters after modification, exit without saving the modified value;

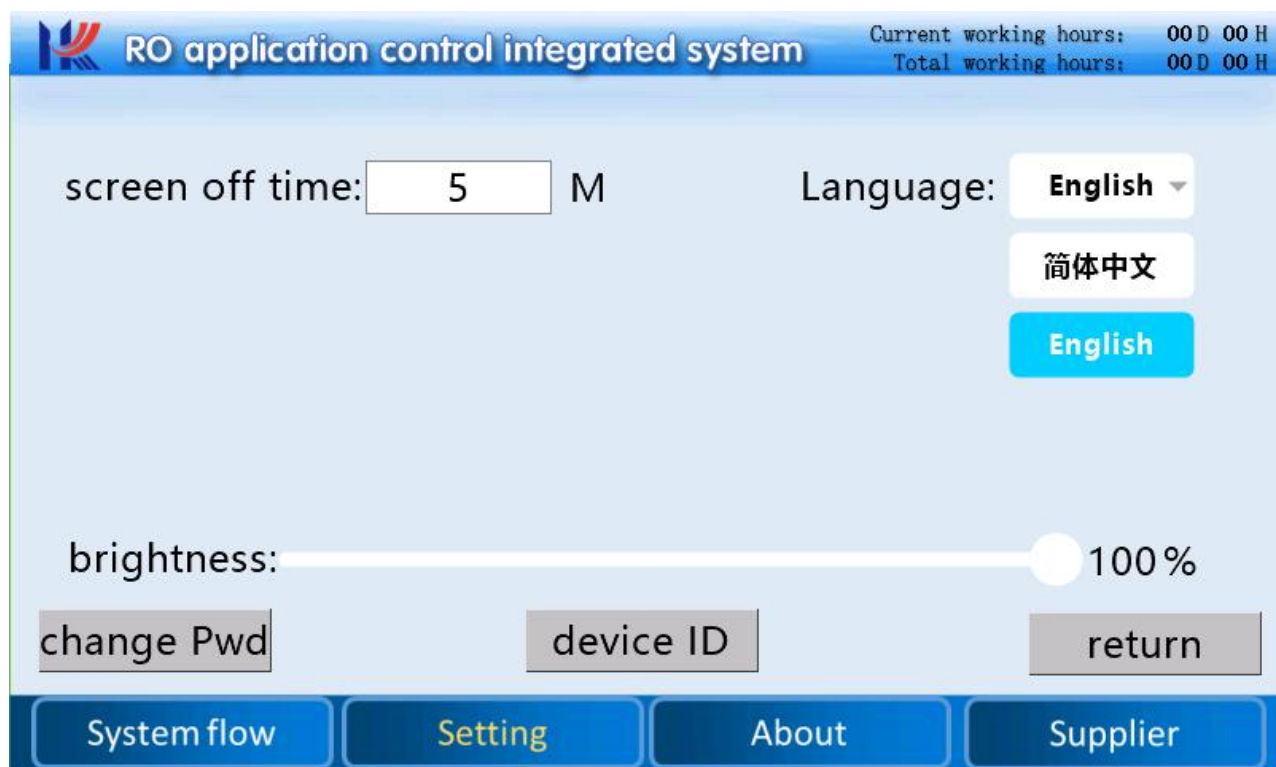
Click the Restore Factory button to automatically modify the running parameters and parameters to the factory default values;

parameter	Factory default	scope	Meaning
flushing time in water making process	30 seconds	0-255	Rinse duration during water production
Flushing interval in water making process	60 minutes	1-255	Water-making process flush interval
Ozone sterilization time	30 seconds	1-255	Ozone sterilization start time
Ozone sterilization interval	120 seconds	1-255	Ozone disinfection interval
Flushing time after full water	30 seconds	0-255	Pure water tank water full flush duration
Flushing interval after full water	2 hours	1-255	Pure water tank water full flush interval time
Timeout time for a single continuous water production	4 hours	1-255	Maximum continuous running time of high pressure pump
Delay time for starting water production	0 minutes	0-255	Pure water level is below the high-level switch delay water-making time

Flush mode	HP flush	HP/LP flush	Reverse osmosis membrane flushing method
Delay time for low pressure detection	10 seconds	1-255	Low voltage switch abnormal waiting time, this time is processed according to the normal low voltage switch
Alarm method when the temperature is too low	alarm	Alarm/stop	Equipment below the set temperature alarm reminder or stop running
Delay time for restarting after device failure	5 minutes	1-255	Abnormal conditions occurred during operation, the equipment paused and delayed startup waiting time.
Heater switch	ON	ON/OFF	Heat preservation heating automatic switch on or off switch
Starting temperature of temperature control heating	5℃	1-255	Automatic start temperature for insulation heating
Temperature control heating stop temperature	10℃	1-255	Heating temperature cutoff, heating will not start if the temperature is less than the heating start temperature when heating stops
Operating mode when power on	memory	memory /manual	The operation mode of the device when power is cut off and restarted again
Delayed opening time of HP pump after low pressure	5 seconds	1-255	When starting water production, after the low pressure switch is normal, the high pressure pump starts waiting time.
Number of retries after low pressure abnormality	3 number	255	The number of retries for low voltage restart when the equipment is stopped due to abnormal conditions

System parameter setting

1. Click the System Settings button in the running interface to enter the settings interface;
2. The range and description of system parameter settings are shown in the figure;

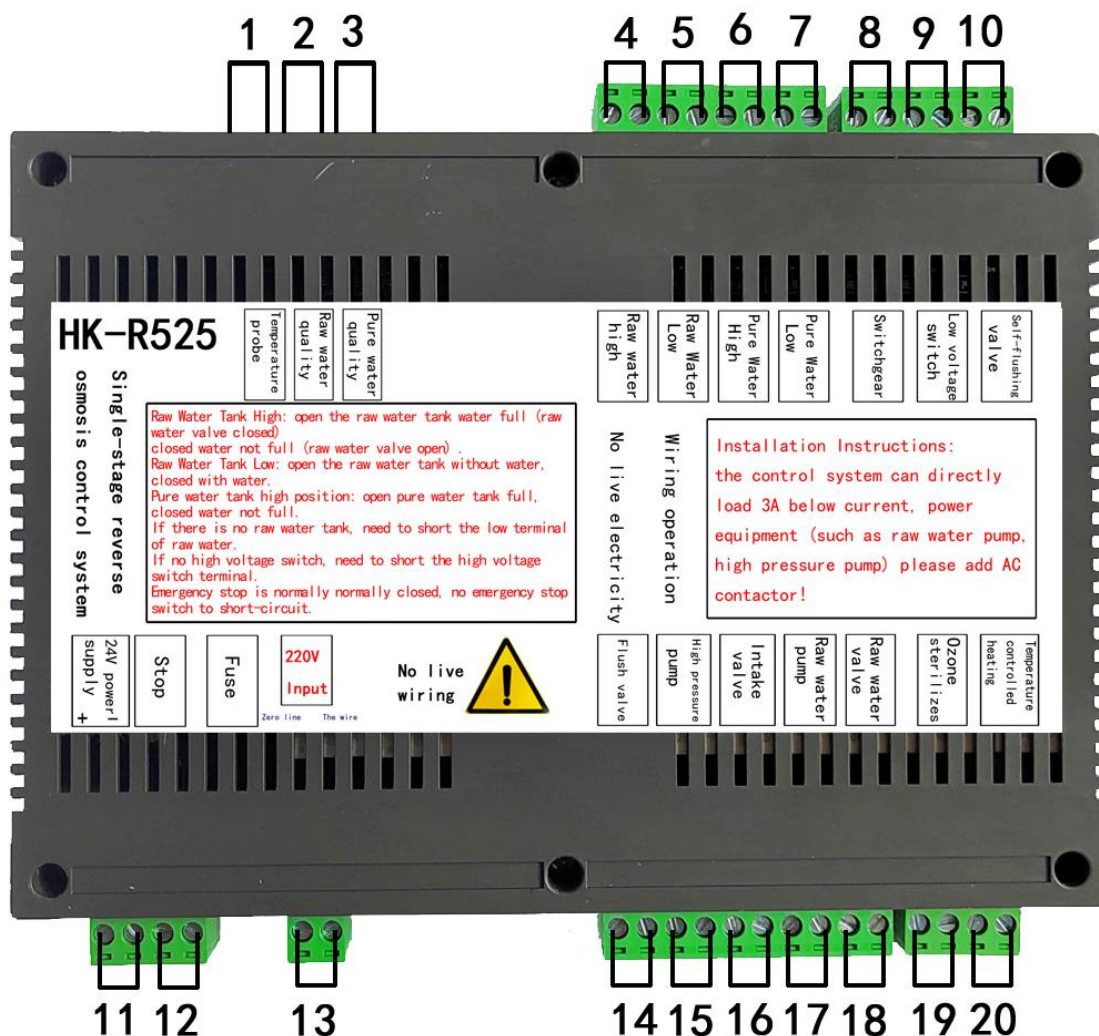


parameter	Factory default	scope	Meaning
Screen off time	5 分钟	1-255	Screen activation display brightness duration, switch to screen saver brightness after timeout
language	English	English / 简体中文	Switch to English interface and save, then restart the device.
brightness	100	30-100	Slider left and right to adjust screen brightness
Change Pwd	0000		Default password 0000, can modify password
Device ID			Device QR code, no operation required.

Default process

1. Enter the setting parameter interface, if no operation is performed within 3 minutes, the system will automatically exit and jump to the running interface;
2. Long press the function key for 2 seconds to restart the display, press the key within 10 seconds after releasing the key is invalid;
3. If the probe is not connected, the temperature value will not be displayed in the running interface, when the insulation heating is started, it will be automatically turned off after continuous heating for 1 minutes and then turned on again, when the set start temperature value is less than the stop temperature, the insulation heating will not start;
4. After low pressure abnormality water making overtime, the equipment will be suspended, and the equipment will be automatically started after waiting for the abnormal start delay time;
5. The raw water tank valve will close after being continuously opened for 4 hours, and will be reopened after waiting for the abnormal start delay time again;

Connection port description



1. Temperature probe: Temperature detection port, a temperature probe is required to be connected when insulation control is needed;
2. Raw water probe: Conductivity detection port for raw water;
3. Pure water probe: Conductivity detection port for pure water;
4. Raw water high level: High level detection of raw water tank, the water tank is disconnected when full, and closed when not full, and the raw water valve is opened after closing;
5. Raw water level: Low water level detection of raw water tank, the water tank is disconnected when empty, and closed when there is water, and the raw water

pump is started after closing;

6. Pure water high level: High water level switch of pure water tank, disconnected when full, closed when not full, and the water production is started when closed;
7. Pure water low level: Low water level switch of pure water tank, closed when there is water, and disconnected when empty;
8. High pressure switch: High water pressure detection the reverse osmosis membrane, disconnected when the water pressure is high, closed when the water pressure is normal, and the water production is stopped when disconnected to prevent high water pressure damaging the reverse osmosis membrane, (if there is no high pressure switch, this port needs to be short-circuited);
9. Low pressure switch: Water detection before the high pressure pump, disconnected when there is no water pressure, closed when the water pressure is normal, and the water production is started when closed;
10. flush valve: Self-flush valve head detection, disconnected when self-flush is closed, closed when started, and the raw water pump is started for self-flush operation after closing;
11. 24V power supply: Controller power supply;
12. Emergency stop: Emergency stop switch connection port, short-circuit when there no emergency stop switch;
13. 220V input: Control end power input, the wiring sequence is left zero and right fire, and the control end output is 220V;
14. Flush valve: Flush valve control output, started when flushing at startup;

15. High pressure pump: pressure pump control output, external AC contactor controls the start of the high pressure pump;
16. Water inlet valve: Water inlet valve control output, started when the raw pump is started and there is no self-flush;
17. Raw water pump: Raw water pump control output, external AC contactor controls the start of the raw water;
18. Raw water valve: Raw water valve control output, started when the raw water tank is high;
19. Ozone sterilization: Ozone sterilization control output, run automatically according to the set sterilization time and sterilization interval;
20. Temperature control heating: Insulation control output, a temperature probe is required to be connected, and the insulation control does not start without connecting a temperature control probe

HK-525 Single-stage reverse osmosis control system

